

**Economica Obscura:
Institutions, Archives, and Policy Learning
Online Appendix**

Chae-Yeon Xon

September 2026

Abstract

This appendix contains proofs, posterior calculations, data construction, and predictive diagnostics for the paper. The first sections develop the institutional control problem, learning limits, reporting thresholds, state persistence, and content-contingent measurement. The remaining sections document the U.S. bank posterior, forecast construction, allocation rules, reform cohorts, and additional predictive diagnostics.

Appendix A. Target Information and Operating Directions

A.1. Fresh-Observation Information

For a differentiable target with nonzero gradient a , define

$$(A1) \quad b_{\mathcal{J}}(a) = \begin{cases} a' \mathcal{J}^+ a, & a \in \text{Range}(\mathcal{J}), \\ +\infty, & a \notin \text{Range}(\mathcal{J}), \end{cases} \quad \ell_{\mathcal{J}}(a) = b_{\mathcal{J}}(a)^{-1}.$$

The conventions are $1/(+\infty) = 0$ and $\mathcal{J}^+$ equal to the Moore–Penrose inverse.

LEMMA A1 (Target information). *For $\mathcal{J} \geq 0$ and $a \neq 0$, $\ell_{\mathcal{J}}(a) = \inf_{a'v=1} v' \mathcal{J} v$. This quantity and $b_{\mathcal{J}}(a)$ are invariant under an invertible change of parameter coordinates with the corresponding transformation of a .*

For a unit target $e(\varphi)$ and $\lambda_L > 0$,

$$(A2) \quad b_{\varphi}(\delta) = \frac{1}{g^2} \left\{ \frac{\cos^2(\varphi - \delta)}{\lambda_H} + \frac{\sin^2(\varphi - \delta)}{\lambda_L} \right\}, \quad \frac{1}{g^2 \lambda_H} \leq b_{\varphi}(\delta) \leq \frac{1}{g^2 \lambda_L}.$$

For d orthonormal targets, the set of such costs has a matrix characterization. Write $b < z$ when the sums of the largest k coordinates of b are bounded by those of z for $k < d$ and their total sums agree.

PROPOSITION A1 (Fresh-observation allocation). *Let $\lambda_1 \geq \dots \geq \lambda_d > 0$, $\mathcal{J}_R = g^2 R' \wedge R$, and let A be an orthogonal matrix whose columns are target gradients. As R ranges over the orthogonal group, the attainable vectors $\text{diag}(A' \mathcal{J}_R^{-1} A)$ form*

$$(A3) \quad \mathcal{B}(\wedge) = \{b : b < g^{-2}(\lambda_d^{-1}, \dots, \lambda_1^{-1})'\} = \text{conv}\{\pi(g^{-2}\lambda^{-1}) : \pi \text{ is a permutation}\}.$$

For target-demand eigenvalues $d_1 \geq \dots \geq d_d \geq 0$, the minimum of $\text{tr}(D \mathcal{J}_R^{-1})$ is $g^{-2} \sum_j d_j / \lambda_j$.

The set in Proposition A1 is the Schur–Horn set of the inverse information matrix. Its trace is a fresh-observation accounting identity. Economic restrictions on the realized orientation enter through the decision problems in the next section. The pure target-demand minimization in the proposition sets operating and adjustment costs aside. A planner with positive operating costs retains those costs when its adjustment cost tends to zero.

A.2. Proof of Lemma A1

Suppose first that $a \in \text{Range}(\mathcal{J})$. Decompose $v = v_R + v_N$ into the information range and kernel. The constraint is $a'v_R = 1$. On the range, Cauchy–Schwarz gives

$$(A4) \quad 1 = (a'v_R)^2 \leq (a'\mathcal{J}^+a)(v_R'\mathcal{J}v_R).$$

Equality is attained at $v_R = \mathcal{J}^+a/(a'\mathcal{J}^+a)$. Thus the infimum equals $(a'\mathcal{J}^+a)^{-1}$. If a has a nonzero kernel projection a_N , the vector $v = a_N/\|a_N\|^2$ satisfies $a'v = 1$ and $v'\mathcal{J}v = 0$. This proves the range convention and the variational formula.

Let $\theta = T\vartheta$ with invertible T . The transformed information and target gradient are $\mathcal{J}_\vartheta = T'\mathcal{J}T$ and $a_\vartheta = T'a$. The substitution $v_\theta = Tv_\vartheta$ is a bijection between the two feasible tangent sets and preserves the quadratic form. The variational infima agree, so both target information and its inverse agree, including the zero and infinite cases.

A.3. Units and a Declared Metric

The raw eigenvalues of information depend on parameter units. For example, $\text{diag}(4, 1)$ and $\text{diag}(1, 4)$ have the same eigenvalues. Under $\theta = \text{diag}(2, 1)\vartheta$, the transformed matrices are $\text{diag}(16, 1)$ and $\text{diag}(4, 4)$. Lemma A1 applies to the corresponding transformed gradients in each experiment.

A positive-definite tangent metric H gives an alternative way to fix the comparison. Use the generalized eigenvalues in $\mathcal{J}v = \lambda Hv$. Under the coordinate change, $H_\vartheta = T'HT$, and the generalized eigenvalues are preserved. Equivalently, use the eigenvalues of $H^{-1/2}\mathcal{J}H^{-1/2}$. Proposition A1 then applies in the whitened coordinates to a target basis satisfying $a_i'H^{-1}a_j = \mathbf{1}\{i = j\}$, provided the maintained information family has the specified orthogonal orbit in those coordinates. The paper uses the declared economic units and $H = I$.

A.4. Operating Choices

Let $B = LSN'$ be a singular value decomposition. For an orthogonal V , $L'VN$ is orthogonal, and

$$(A5) \quad \text{tr}(B'V) = \text{tr}(SL'VN) \leq \sum_j S_{jj}.$$

The value is attained at $V = LN'$. If all singular values are positive, equality requires every diagonal entry of $L'VN$ to equal one, which implies $L'VN = I$. Hence the maximizer is unique and equals $B(B'B)^{-1/2}$.

Write $C = L_C \text{diag}(c_1, \dots, c_d) L_C'$ with $c_1 \geq \dots \geq c_d$. For $Z = L_C' U$, the entries Z_{ij}^2 form a doubly stochastic matrix. The objective equals $\sum_{ij} c_i \lambda_j Z_{ij}^2$. The rearrangement inequality bounds it above by $\sum_j c_j \lambda_j$, attained by ordered eigenvector alignment. Repeated eigenvalues allow rotations within the corresponding eigenspaces. These static choices are conditional on the moments M_r, C_r ; restrictions on their empirical variation must be supplied by the application.

A.5. Proof of Proposition A1

The inverse matrix is $\mathcal{J}_R^{-1} = g^{-2} R' \Lambda^{-1} R$. As R ranges over the orthogonal group, RA also ranges over that group. The real symmetric Schur–Horn theorem states that the diagonals of the orthogonal orbit of a matrix with ordered eigenvalues z are exactly the vectors majorized by z . This gives (A3), with $z = g^{-2}(\lambda_d^{-1}, \dots, \lambda_1^{-1})'$. The convex-hull representation is the equivalent permutation characterization of majorization.

Diagonalize $D = L_D \text{diag}(d_1, \dots, d_d) L_D'$. The trace $\text{tr}(D \mathcal{J}_R^{-1})$ is minimized by pairing the largest d_j with the smallest inverse-information eigenvalue $1/(g^2 \lambda_j)$. This yields the stated minimum. Pairing in the reverse order gives the maximum $g^{-2} \sum_j d_j / \lambda_{d+1-j}$.

In the rank-deficient case, the positive-information subspace rotates with the orthogonal basis. A target has finite fresh-observation cost precisely when its gradient belongs to that subspace. This condition uses all free components of the parameter. For example, consider

$$(A6) \quad Y = \cos \delta \theta_1 + \sin \delta \theta_2 + \varepsilon, \quad \varepsilon \sim N(0, 1).$$

For target θ_1 and $\sin \delta \neq 0$, $v = (1, -\cot \delta)'$ satisfies the target constraint and lies in the information kernel. Lemma A1 gives target information zero. If θ_2 is fixed and known, the scalar experiment has information $\cos^2 \delta$ for θ_1 . The scalar experiment in Appendix E specifies its parameter dimension.

Appendix B. The Institutional Control Problem

B.1. Proof of Lemma 1: Quadratic Value and the Priority Sequence

For $0 < \eta < 1$, write the date- t value for one institution as

$$(A7) \quad V_t(x) = \frac{1}{2} P x^2 - r_t x + s_t.$$

The Bellman problem is

$$(A8) \quad V_t(x) = \min_y \left\{ \frac{\rho}{2} (y - \bar{x}_t)^2 + \frac{\kappa}{2} (y - x)^2 + \eta V_{t+1}(y) \right\}.$$

Let $H_0 = \rho + \kappa + \eta P$. Completing the square yields

$$(A9) \quad y_t(x) = \frac{\kappa x + \rho \bar{x}_t + \eta r_{t+1}}{H_0}, \quad P = \kappa - \frac{\kappa^2}{H_0}, \quad r_t = \frac{\kappa}{H_0}(\rho \bar{x}_t + \eta r_{t+1}).$$

The coefficient equation is the Riccati equation in (6). Its positive solution is the stated root of

$$(A10) \quad \eta P^2 + [\rho + \kappa(1 - \eta)]P - \kappa\rho = 0.$$

It gives $0 < \gamma = \kappa/H_0 < 1$ and $P = \kappa(1 - \gamma)$. Bounded priorities and $\eta\gamma < 1$ give the bounded solution

$$(A11) \quad r_t = \gamma\rho \sum_{j=0}^{\infty} (\eta\gamma)^j \bar{x}_{t+j}.$$

The constant term satisfies $s_t = \rho \bar{x}_t^2/2 + \eta s_{t+1} - (\rho \bar{x}_t + \eta r_{t+1})^2/(2H_0)$ and has a bounded solution obtained by discounted summation. Substituting (A11) into the optimizer gives

$$(A12) \quad y_t(x) = \gamma x + \frac{\rho}{H_0} \sum_{j=0}^{\infty} (\eta\gamma)^j \bar{x}_{t+j}.$$

The identity $\rho/H_0 = (1 - \gamma)(1 - \eta\gamma)$ transforms this expression into (7).

B.2. Verification and Uniqueness

For a bounded installed path, iterating the completed-square Bellman identity and letting the terminal date grow gives its total cost as $V_0(x_0)$ plus

$$(A13) \quad \frac{H_0}{2} \sum_{t=0}^{\infty} \eta^t [x_{t+1} - y_t(x_t)]^2.$$

The same identity applies to every finite-cost path: bounded priorities and the operating-cost term imply $\sum_t \eta^t x_{t+1}^2 < \infty$, hence the discounted quadratic terminal value tends to zero. The path in (7) is bounded and makes every square zero. The sum of squares is positive for any path differing at a finite date. This proves optimality and uniqueness for $0 < \eta < 1$. The per-date one-period problem gives $\gamma = \kappa/(\rho + \kappa)$ when continuation is assigned zero weight.

The two institutions share the same γ and averaging weights. Taking the difference of their policy functions gives the relative-direction law. Under a constant priority, iteration

gives $\delta_t - \bar{\delta} = \gamma^t(\delta_0 - \bar{\delta})$. Finally,

$$(A14) \quad \frac{\rho}{\kappa} = \frac{(1-\gamma)(1-\eta\gamma)}{\gamma} = \frac{1}{\gamma} - (1+\eta) + \eta\gamma.$$

The right side has derivative $-1/\gamma^2 + \eta < 0$. Increasing κ therefore increases γ . This establishes the comparative static in Lemma 1.

B.3. Unequal Coefficients and Matrix Installation

For institution $j \in \{A, O\}$, let $(\rho_j, \kappa_j, \eta_j)$ define its own Riccati solution and persistence γ_j . Its policy is (7) with those coefficients. The difference $\beta_{t+1} - \alpha_{t+1}$ depends on both inherited angles whenever the persistence coefficients differ. A state containing both installed angles then supplies the required inherited information.

For matrix-valued directions, installed bases can be updated by $V_{t+1} = V_t \exp(A_t)$ and $U_{t+1} = U_t \exp(B_t)$, where A_t, B_t are skew symmetric. Their relative orientation satisfies

$$(A15) \quad R_{t+1} = \exp(-B_t) R_t \exp(A_t).$$

This identity records the effect of two installation choices. The solved control problem in Lemma 1 uses the scalar angular coordinate and the quadratic objective (5).

Appendix C. Posterior Loss and the Learning Limit

C.1. Proof of Lemma 2

Conditional Gaussian likelihoods add their quadratic terms to the log prior. Their Hessians sum to J_T in (8), giving posterior covariance J_T^{-1} . Completing the policy-loss square gives

$$(A16) \quad L(q, \theta; a, c) - L(q^*(\theta), \theta; a, c) = \frac{c}{2} \left(q - \frac{a'\theta}{c} \right)^2.$$

Conditional minimization gives $\hat{q} = a'\mu_T/c$. Its conditional expected loss is $a'J_T^{-1}a/(2c)$. Averaging over the independent policy problem gives (9). For fixed $J > 0$,

$$(A17) \quad (J_0 + nJ)^{-1} = \frac{1}{n}J^{-1} - \frac{1}{n^2}J^{-1}J_0J^{-1} + O(n^{-3}).$$

The trace expansion proves the stated approximation. The smallest eigenvalue of the fixed matrix J is positive throughout this calculation.

C.2. Proof of Theorem 1

In the orthonormal basis (u, v) , put $z_s = \gamma^s \varepsilon_0$ and write

$$(A18) \quad J_T = \begin{pmatrix} A_T & B_T \\ B_T & C_T \end{pmatrix}, \quad A_T = u' J_0 u + \tau \sum_{s < T} \cos^2 z_s, \\ B_T = u' J_0 v + \tau \sum_{s < T} \cos z_s \sin z_s, \quad C_T = v' J_0 v + \tau \sum_{s < T} \sin^2 z_s.$$

The bounds $|\sin z_s| \leq |z_s|$ and $|\cos z_s| \leq 1$ imply

$$(A19) \quad \sum_{s=0}^{\infty} \sin^2 z_s \leq \frac{\varepsilon_0^2}{1 - \gamma^2}, \quad \sum_{s=0}^{\infty} |\cos z_s \sin z_s| \leq \frac{|\varepsilon_0|}{1 - \gamma}.$$

Thus $C_T \rightarrow C_\infty > 0$, B_T has a finite limit, and $A_T = \tau T + O(1)$. Let $S_T = C_T - B_T^2/A_T$. Then $S_T \rightarrow C_\infty$, and block inversion gives

$$(A20) \quad J_T^{-1} = \begin{pmatrix} A_T^{-1} + B_T^2/(A_T^2 S_T) & -B_T/(A_T S_T) \\ -B_T/(A_T S_T) & S_T^{-1} \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 0 \\ 0 & C_\infty^{-1} \end{pmatrix}.$$

Returning to the original coordinates gives (11). Taking the trace against $D/2$ gives the limiting policy loss. If $0 < |\varepsilon_0| \leq \pi/2$, each term with $s \geq 1$ strictly increases with γ because $\sin^2 z$ increases with $|z|$ for $|z| \in (0, \pi/2)$. The sum is therefore strictly increasing. Combining this with (A14) proves the statement about adjustment costs.

C.3. Proof of Proposition 1

Substituting $\delta_t - \bar{\delta} = \gamma^t \varepsilon_0$ into the relative operating and adjustment costs and summing their discounted values gives (12). Its derivative has the sign of $\rho\gamma - \kappa(1 - \gamma)(1 - \eta\gamma)$. This expression is strictly increasing from $-\kappa$ to ρ , so it has one root $\gamma^E \in (0, 1)$, and $\mathcal{C}^G$ decreases below that root and increases above it. Its second derivative is positive at γ^E .

Theorem 1 makes $C_\infty(\gamma)$ strictly increasing on $(0, 1)$, so $\mathcal{L}^G$ is strictly decreasing there. As $\gamma \uparrow 1$, the first M terms in (10) converge to $M \sin^2(\varepsilon_0)$ for each fixed M ; letting M grow gives $C_\infty(\gamma) \rightarrow \infty$. The extensions of $\mathcal{C}^G$ and $\mathcal{L}^G$ are therefore continuous at one. On $[0, \gamma^E]$, the social objective decreases strictly, including at γ^E through the policy-loss derivative. Compactness then gives a global minimizer in $(\gamma^E, 1]$.

At $\chi = 0$, the unique global minimizer is the interior point γ^E . Continuity of the objective confines every minimizer to a neighborhood of γ^E for sufficiently small χ . The implicit-function theorem applied to $\mathcal{C}^{G'}(\gamma) + \chi \mathcal{L}^{G'}(\gamma) = 0$ supplies a unique local minimizer in that neighborhood. Since $\mathcal{L}^{G'}(\gamma^E) = -\bar{d}C'_\infty(\gamma^E)/[2C_\infty(\gamma^E)^2]$, differentiation at $\chi = 0$ gives (14). Positivity follows from $C'_\infty(\gamma^E) > 0$ and $\mathcal{C}^{G''}(\gamma^E) > 0$.

C.4. Proof of Corollary 1

With $\lambda_L > 0$, each batch satisfies $n\mathcal{J}(\delta_s) \geq ng^2\lambda_L I_2$. Therefore $J_T \geq Tng^2\lambda_L I_2$ and $J_T^{-1} \leq (Tng^2\lambda_L)^{-1}I_2$. Taking the trace against $D/2$ proves the bound.

In the rank-one case, the supplemental observations add ξm_T to C_T and leave A_T, B_T unchanged in (A18). The Schur complement is $S_T = C_T + \xi m_T - B_T^2/A_T$. If $m_T \rightarrow \infty$, $S_T \rightarrow \infty$ and $A_T \rightarrow \infty$, while B_T stays bounded. Every entry of the inverse in (A20) then tends to zero. This argument applies to every diverging sequence m_T , including sequences with $m_T/T \rightarrow 0$.

C.5. Precision, Rank, and Policy Targets

For positive semidefinite information matrices I_s ,

$$(A21) \quad \ker\left(\sum_s I_s\right) = \bigcap_s \ker(I_s).$$

Indeed, $v'(\sum_s I_s)v = \sum_s v'I_s v$ vanishes precisely when every nonnegative summand vanishes. This proves the finite-archive rank statement in the main text. Posterior concentration in every direction requires the smallest eigenvalue of J_T to diverge. Theorem 1 computes that eigenvalue's finite limit under the specified path.

For a restricted set of future targets, the posterior loss is $\text{tr}(DJ_T^{-1})/2$. In Theorem 1, its limit is zero exactly when $v'Dv = 0$. Because $D \geq 0$, this condition is equivalent to future target gradients lying in u 's span with probability one under their loss weights. The policy consequence of residual parameter uncertainty thus depends on the target distribution.

Appendix D. The Planner and Reporting Threshold

D.1. State Reduction and Existence

Let $\alpha = m - \delta/2$ and $\beta = m + \delta/2$, and define the corresponding common and relative desired positions. The sum of operating costs is $\rho(m_{t+1} - \bar{m}_t)^2 + \rho(\delta_{t+1} - \bar{\delta}_t)^2/4$. The sum of adjustment costs is $\kappa(m_{t+1} - m_t)^2 + \kappa(\delta_{t+1} - \delta_t)^2/4$. Information in (4) depends on the relative angle. Minimizing over the common path therefore adds a term independent of all relative choices and yields (21).

For finite H , the value recursion starts with the continuous zero function. At any positive-definite J , policy risk is finite, continuous, and nonnegative. By induction, the continuation value is continuous and nonnegative. On compact sets of states (x, J) with $J > 0$, the quadratic term $\kappa(y - x)^2/4$ supplies a uniform bound on minimizing choices, using the feasible choice $y = x$ as a locally bounded upper bound for value. Minimization

over that compact set gives a continuous value and an attained minimum. This establishes existence by backward induction.

For $J_2 \geq J_1 > 0$, any fixed future angle sequence has weakly smaller policy losses starting from J_2 . The sequence's operating costs are unchanged. Taking the infimum over sequences gives $\mathcal{V}_t(x, J_2) \leq \mathcal{V}_t(x, J_1)$. Where the matrix gradient exists under the trace inner product, it is negative semidefinite. Hence $M_t \geq 0$ in (23).

D.2. The Marginal Reporting Condition

At a differentiability point of the continuation value, perturb y and write $J^+ = J + n_t \mathcal{J}(y)$. The derivative of inverse precision is $-n_t(J^+)^{-1} \mathcal{J}'(y)(J^+)^{-1}$. The chain rule applied to (22) yields

$$(A22) \quad \begin{aligned} 0 = & \frac{\rho}{2}(y - \bar{\delta}_t) + \frac{\kappa}{2}(y - x) - \frac{\chi n_t}{2} \text{tr}\{(J^+)^{-1} D_t(J^+)^{-1} \mathcal{J}'(y)\} \\ & + \eta \partial_x \mathcal{V}_{t+1}(y, J^+) + \eta n_t \text{tr}\{\nabla_J \mathcal{V}_{t+1}(y, J^+) \mathcal{J}'(y)\}. \end{aligned}$$

Collecting the two information derivatives gives (24). At a state with multiple optimal continuation branches, the Bellman minimization remains the definition of value; the displayed derivative condition applies to smooth branches where the indicated derivatives exist.

D.3. Proof of Theorem 4

Let $A = \text{diag}(a_0, b_0)$. The posterior precision is $A + \tau e(\delta) e(\delta)'$. Its trace is N and its determinant is

$$(A23) \quad \det[A + \tau e(\delta) e(\delta)'] = a_0 b_0 + \tau \{b_0 \cos^2 \delta + a_0 \sin^2 \delta\} = K + L \sin^2 \delta.$$

For a two-by-two positive-definite matrix, the inverse trace is its trace divided by its determinant. This gives (37). If $a_0 = b_0$, then $L = 0$ and the information term is constant, so the quadratic cost selects zero.

Suppose $a_0 > b_0$. Then $L > 0$. From (40),

$$(A24) \quad F(\delta) - F(0) \geq \left(\frac{h}{4} - \frac{\chi NL}{2K^2} \right) \delta^2.$$

The bound follows by $K + L \sin^2 \delta \geq K$ and $\sin^2 \delta \leq \delta^2$. It is strict for a nonzero δ at $\chi = \chi^*$ because $\sin^2 \delta < \delta^2$. Thus zero is the unique global minimum up to and including the threshold.

For $\chi > \chi^*$, $F''(0) = h/2 - \chi NL/K^2 < 0$. For $0 < \delta < \pi/2$, a stationary point solves $\chi = \Xi(\delta)$, where

$$(A25) \quad \Xi(\delta) = \frac{h\delta[K + L \sin^2 \delta]^2}{NL \sin(2\delta)}.$$

The function $\delta/\sin(2\delta)$ strictly increases on this interval: its derivative has the sign of $\sin(2\delta) - 2\delta \cos(2\delta)$, whose derivative is $4\delta \sin(2\delta) > 0$ and whose limit at zero is zero. The factor $[K + L \sin^2 \delta]^2$ is positive and strictly increasing. Therefore Ξ strictly increases, with limits χ^* at zero and $+\infty$ at $\pi/2$. There is exactly one positive stationary point, the objective decreases before it and increases after it, and reflection gives the negative minimizer. Periodicity of the information term and the displacement cost exclude every representative with larger absolute angle. This proves the global characterization and the limits of δ^* .

To differentiate the threshold, set $z = a_0 - b_0 > 0$, $d = b_0 + \tau$, and $e = 2b_0 + \tau$. Then

$$(A26) \quad \chi^* = \frac{hb_0^2}{2\tau} \frac{(z+d)^2}{z(z+e)}, \quad \frac{d}{dz} \log \frac{(z+d)^2}{z(z+e)} = \frac{-\tau z - de}{(z+d)z(z+e)} < 0.$$

This proves the derivative and both limits in the theorem. An archive $J = j_2 I_2 + p e_1 e'_1$ with $p \geq 0$ provides the comparison directly: increasing the precision p of retained first-coordinate reports raises a_0 while leaving b_0 fixed.

D.4. Proof of Corollary 2

With a constant zero priority, the institution's value at an installed angle x is $Px^2/2$. After the planner chooses (α', β') , discounted operating continuation is $\eta P(\alpha'^2 + \beta'^2)/2$. Write $m' = (\alpha' + \beta')/2$ and $\delta = \beta' - \alpha'$. Starting from zero and adding the current operating and adjustment costs gives $h_c m'^2 + h_c \delta^2/4$, where $h_c = \rho + \kappa + \eta P$. The common direction minimizes at zero. The policy loss of the next report is unchanged, so the objective is (37) with curvature h_c . Theorem 4 applies with that substitution.

D.5. The Terminal Local Change

Let $F_\chi(\delta) = h(\delta - \delta^E)^2/4 + \chi r(\delta)$. Since inherited $J > 0$, r is smooth. The first-order condition is $h(\delta - \delta^E)/2 + \chi r'(\delta) = 0$. At $(\delta, \chi) = (\delta^E, 0)$ its derivative with respect to δ is $h/2 > 0$. The implicit function theorem gives $\delta^{SP} - \delta^E = -2\chi r'(\delta^E)/h + O(\chi^2)$. Substituting this into a third-order expansion of F_χ gives the objective gain in (41). The positive cost curvature makes the stationary branch a local minimum for sufficiently small χ .

Appendix E. A Scalar Experiment at an Information Boundary

This appendix uses one free structural parameter $\theta \in \mathbb{R}$ and a two-dimensional response. Let $x = ge(\alpha)\theta$ and observe

$$(A27) \quad Y = \sqrt{\lambda} e(\beta)' x + \varepsilon = g\sqrt{\lambda} \cos(\beta - \alpha)\theta + \varepsilon, \quad \varepsilon \sim N(0, 1).$$

The response derivative is $G = ge(\alpha) \in \mathbb{R}^{2 \times 1}$. Measurement information is $K = \lambda e(\beta)e(\beta)'$. The scalar information is $\mathcal{J}(r) = \lambda g^2 \cos^2 \delta(r)$. The response singular value and the positive measurement eigenvalue are fixed. The scalar information varies with their relative direction.

Suppose an announced regime r is expected to persist. The inherited direction is held fixed, and the preferred relative direction is $\bar{\delta}(r)$. Lemma 1 gives the next direction as $\delta(r) = \gamma \delta_{\text{old}} + (1 - \gamma) \bar{\delta}(r)$. Let $\delta(r^*) \in \pi/2 + \pi\mathbb{Z}$, and suppose the first nonzero derivative of $\bar{\delta}$ at r^* has order $m \geq 1$. Taylor expansion gives

$$(A28) \quad \mathcal{J}(r) = C_m |r - r^*|^{2m} + o(|r - r^*|^{2m}), \quad C_m = \lambda g^2 (1 - \gamma)^2 \left[\frac{\bar{\delta}^{(m)}(r^*)}{m!} \right]^2.$$

For $r_n - r^* = zn^{-1/(2m)}$, the total information from n fresh observations tends to $C_m z^{2m}$. This is a rate at which the regime approaches the boundary. With a scalar Gaussian prior of precision $j_0 > 0$, posterior variance along this sequence tends to $(j_0 + C_m z^{2m})^{-1}$. The order m is an assumption on the regime-priority schedule in this experiment.

The expansion follows from $\delta(r) - \delta(r^*) = (1 - \gamma) \bar{\delta}^{(m)}(r^*) (r - r^*)^m / m! + o(|r - r^*|^m)$ and $\cos^2(\pi/2 + z) = z^2 + o(z^2)$. It concerns the scalar parameter specified in (A27). The rank-one two-parameter target in (A6) uses the range condition in Lemma A1. In a nonlinear experiment, zero local Fisher information can also coexist with global injectivity; $Y \sim N(\theta^3, 1)$ at $\theta = 0$ is one example.

E.1. Projection Inference

Let an empirical moment vector $\widehat{m}_n$ have mean $m(\theta, r) \in \mathbb{R}^q$. Assume a uniform central limit theorem over the maintained parameter-regime class and uniform consistency of a covariance estimator $\widehat{\Omega}_n$, with covariance eigenvalues bounded above and bounded away from zero. Define

$$(A29) \quad Q_n(\theta, r) = n[\widehat{m}_n - m(\theta, r)]' \widehat{\Omega}_n^{-1} [\widehat{m}_n - m(\theta, r)], \quad \mathcal{C}_n(r) = \{\theta : Q_n(\theta, r) \leq \chi_{q, 1-\alpha}^2\}.$$

At the true parameter, the uniform central limit theorem and covariance consistency imply a uniform χ_q^2 limit for Q_n . Hence $\liminf_n \inf_{(\theta, r)} P_{\theta, r}\{\theta \in \mathcal{C}_n(r)\} \geq 1 - \alpha$. Projection

gives the same lower coverage bound for $\{\psi(\vartheta) : \vartheta \in \mathcal{C}_n(r)\}$. For Gaussian sample means with known covariance, the true-parameter statistic has its chi-square distribution at each finite n . These assumptions concern moment distributions and covariance estimation; they allow the derivative of m with respect to θ to approach zero along the scalar regime sequence.

Appendix F. Record Labels and Observation Maps

F.1. Retaining the Observation Map

Let L be a recorded regime label with a probability distribution independent of θ . Let $s_L(Y; \theta)$ be the conditional score and assume the differentiability and integrability needed to interchange differentiation and integration. Integrating over the label gives score $E[s_L(Y; \theta) | Y]$. The matrix law of total variance gives

$$(A30) \quad \mathcal{I}_{Y,L} - \mathcal{I}_Y = E[\text{Var}\{s_L(Y; \theta) | Y\}] \geq 0.$$

The labelled archive in (8) keeps the observation maps associated with each report. An archive with missing maps requires the corresponding observed-data likelihood. Equation (A30) describes the information difference for the stated mixture experiment.

Appendix G. Cumulative Excitation

G.1. Proof of Lemma 3

Let $S_T = \sum_{s \leq T} n_s \mathcal{I}(\delta_s) \geq 0$. For a positive-definite matrix, $\|J_T^{-1}\|_{\text{op}} = 1/\lambda_{\min}(J_T)$. Weyl's inequalities give $\lambda_{\min}(S_T) + \lambda_{\min}(J_0) \leq \lambda_{\min}(J_T) \leq \lambda_{\min}(S_T) + \lambda_{\max}(J_0)$. Thus the smallest eigenvalue of J_T diverges exactly when that of S_T does. Disjoint blocks of length L under the displayed finite-window condition give $S_T \geq \lfloor T/L \rfloor \epsilon I_d$, proving the rate in Lemma 3. This is a covariance-concentration statement for the specified Gaussian posterior.

Appendix H. Sequential Reporting and Record Value

H.1. Proof of Theorem 5

For two-dimensional vectors, the determinant expansion of $J_0 + \tau \sum_s e(\delta_s) e(\delta_s)'$ is the sum of the initial determinant, the cross-products with J_0 , and the squared areas spanned by pairs of new vectors. The cross-products equal $\tau \sum_s \{b \cos^2 \delta_s + a \sin^2 \delta_s\}$; each squared area equals $\tau^2 \sin^2(\delta_r - \delta_s)$. This proves (43). The trace equals S_t , so $\text{tr}(J_t^{-1}) = S_t / \det J_t$.

Let $E_t(\delta) = \det J_t - K_t \geq 0$ and let $E_t^{(2)}(\delta)$ replace every squared sine in (43) by its squared argument. Then $0 \leq E_t \leq E_t^{(2)}$, and

$$(A31) \quad F_H(\delta) - F_H(0) = \frac{1}{4}\delta' A_H \delta - \frac{\chi}{2} \sum_{t=1}^H \frac{\eta^{t-1} S_t E_t(\delta)}{K_t [K_t + E_t(\delta)]} \geq \frac{1}{4}\delta' (A_H - 2\chi B_H) \delta.$$

The matrix A_H is positive definite. If $\chi < \chi_H^*$, the last quadratic form is positive for every nonzero δ . At equality, consider a nonzero vector in its null space. It has $\delta' B_H \delta > 0$ and hence $E_t^{(2)} > 0$ for some t . For that t , $E_t/(K_t + E_t) < E_t^{(2)}/K_t$: when $E_t = 0$ the right side is positive, and when $E_t > 0$ the denominator gives strict inequality. Thus the original objective difference is positive. This also establishes uniqueness at the threshold. If $B_H = 0$, information has the same value along every path in the only possible case, $H = 1$ and $a = b$, and the quadratic cost selects zero.

For $\chi > \chi_H^*$, choose z with $z'(A_H - 2\chi B_H)z < 0$. Expansion at ϵz gives $F_H(\epsilon z) - F_H(0) = \epsilon^2 z'(A_H - 2\chi B_H)z/4 + O(\epsilon^4)$, which is negative for sufficiently small positive ϵ . Positive definite quadratic costs make the continuous objective coercive, so a minimizer exists and has a nonzero angle. For $a = b$ and $H \geq 2$, the pair $(1, 2)$ gives a positive quadratic form for a path with $z_1 \neq z_2$. The stated $H = 1$ formula follows by substituting $A_1 = \rho + \kappa$ and $B_1 = (a + b + \tau)\tau(a - b)/[b(a + \tau)]^2$.

H.2. Proof of Corollary 4

With $J_0 = jI_2$, the first posterior inverse trace is independent of δ_1 . At the second date it equals $S/[K + \tau^2 \sin^2(\delta_2 - \delta_1)]$. Put $z = \delta_2 - \delta_1$. Four times the discounted operating and adjustment cost equals

$$(A32) \quad h\delta_1^2 + \eta\rho(\delta_1 + z)^2 + \eta\kappa z^2 = (h + \eta\rho) \left(\delta_1 + \frac{\eta\rho z}{h + \eta\rho} \right)^2 + \eta c z^2.$$

Minimizing the square gives the angles in (47). After removing the first policy loss and dividing the remaining objective by η , the problem is

$$(A33) \quad \min_{z \in \mathbb{R}} \left\{ \frac{c}{4} z^2 + \frac{\chi S}{2(K + \tau^2 \sin^2 z)} \right\}.$$

An information-equivalent representative with $|z| \leq \pi/2$ has the smallest cost. The objective difference from zero is $c z^2/4 - \chi S \tau^2 \sin^2 z/[2K(K + \tau^2 \sin^2 z)]$. The argument of Theorem 4 therefore gives the threshold, the unique positive root above it, the reflection, and the limiting angles. In particular, $\sin(2z)/z$ strictly decreases on $(0, \pi/2)$ and the denominator in (47) increases there. Finally, $\partial c/\partial \kappa = 1 + \eta\rho^2/(h + \eta\rho)^2 > 0$ and $\partial c/\partial \eta = -\rho^2 h/(h + \eta\rho)^2 < 0$. The remaining factors of the threshold are constant in these comparisons.

H.3. Proof of Theorem 6

At ζ_0 , Corollary 4 reduces the objective to the completed square in (A32) and the scalar problem (A33). For $\chi > \chi_2^*$, its two solutions $\pm z^0$ lie in the interior of $(-\pi/2, \pi/2)$ and the derivative in the stationary equation crosses zero once on the positive branch. The scalar second derivative is positive at z^0 . The completed square contributes positive curvature in the remaining direction, so the Hessian of F_2 is positive definite at δ^0 and $-\delta^0$.

The first-order condition $\nabla_\delta F_2(\delta; \zeta) = 0$ is continuously differentiable in angles and positive parameters. The implicit function theorem gives two continuously differentiable stationary branches through the benchmark minima and a Lipschitz bound of the form in (48) after shrinking the parameter neighborhood. Positive-definite Hessians persist in that neighborhood.

Quadratic operating and adjustment costs make every common sublevel set lie in a compact angle rectangle for parameters near ζ_0 . At the benchmark, continuity and uniqueness in Corollary 4 give a positive objective gap between neighborhoods of the two minima and the remainder of this rectangle. Uniform continuity preserves part of that gap for nearby parameters. Every global minimizer therefore belongs to one of the two stationary branches. Since both coordinates of δ^0 are separated from zero, the Lipschitz bound preserves their opposite signs after one final reduction of the neighborhood.

H.4. Numerical Map for Theorem 6

A numerical map varies a/b and d_1/d_2 over 31 logarithmically spaced values from $\exp(-0.55)$ to $\exp(0.55)$, holding $ab = d_1d_2 = 1$. At each of 961 points it minimizes the two-angle objective from 25 initial values on $[-1.35, 1.35]^2$. The reports straddle zero at 97.09 percent of the full grid and every point with both ratios in $[0.8, 1.25]$. Within that rectangle the smaller absolute angle is at least 0.115 radians.

H.5. Proof of Lemma 4

The Gaussian conditioning formula gives $P_t^+ = P_t - P_t s s' P_t / (\nu + s' P_t s)$. Future independent innovations contribute the same covariance under the baseline and augmented archives. The covariance difference equals $A^h P_t s s' P_t (A^h)' / (\nu + s' P_t s)$. Taking one half of its trace against D proves (16). Its numerator equals the squared norm of $D^{1/2} A^h P_t s$, which gives the zero-value condition. The covariance may be singular; the argument only uses positivity of the report's predictive variance.

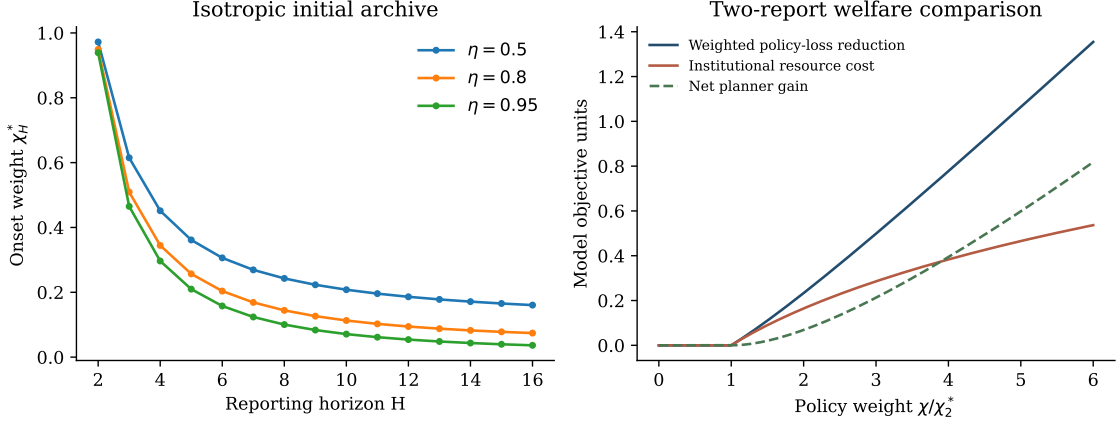


FIGURE A1. Reporting horizon and the welfare wedge

The left panel evaluates the threshold for $a = b = 1$, $\tau = 3$, $\rho = 1$, $\kappa = 2$, $H = 2, \dots, 16$, and the displayed discount factors. These horizon comparisons are numerical. The right panel uses two dates and $\eta = 0.95$ to decompose equation (32). Costs and benefits are measured in model units.

H.6. Proof of Corollary 3

For $B_H \neq 0$, $\chi_H^* = \min_{z: z' B_H z = 1} z' A_H z / 2$. Define $W_\eta = \text{diag}(1, \eta, \dots, \eta^{H-1})$ and the difference operator $(\Delta z)_t = z_t - z_{t-1}$ with $z_0 = 0$. The cost matrices satisfy

$$(A34) \quad A_H = \rho W_\eta + \kappa \Delta' W_\eta \Delta.$$

The square matrix Δ is invertible. Increasing either cost adds a positive-definite matrix E to A_H . The normalization $z' B_H z = 1$ implies $z' E z \geq \lambda_{\min}(E) / \lambda_{\max}(B_H) > 0$, so the minimized quotient strictly increases. Joint scaling of ρ and κ scales A_H and leaves B_H fixed, which proves homogeneity.

Appendix I. A Report-Content Cutoff after a Precision Update

I.1. Proof of Theorem 3

The covariance formulas in (26) follow from conditioning a Gaussian vector on $Y = \theta_1 + \epsilon$. Conditional on $Y = y$, T is Gaussian with mean $\alpha + \lambda k y$ and variance $\sigma_T^2 + \lambda^2 p_1$. This gives $\pi(y)$ and shows that it is strictly increasing in y .

Applying the rank-one covariance update to the second report gives (28). The variance reduction from an e_i report is $A(p_i) e_i e_i'$, where $(p_1, p_2) = (p_1, s_2)$. Hence

$$(A35) \quad R_1(y) - R_2(y) = \frac{1}{2} [d_2 \{\pi(y)\} A_2 - d_1 \{\pi(y)\} A_1] = \frac{1}{2} [d_H A_2 - d_L A_1 - \Delta_d (A_1 + A_2) \pi(y)].$$

The move is selected when $c < \chi\{R_1(y) - R_2(y)\}$. Solving for $\pi(y)$ gives (29). Strict monotonicity of $\pi(y)$ and inversion of the standard normal distribution give (30).

For every realization, the adaptive plan takes the smaller of the two conditional costs. Taking expectations gives a value weakly below the expectation under either fixed action and therefore gives (31). If each action is strictly preferred on an event with positive probability, each fixed action incurs a strictly larger expected cost on one of those events.

Appendix J. Proof for a Changing State

J.1. Proof of Theorem 2

Substituting $A^h = \phi^h I_2$ into (16) gives $\Delta_h = \phi^{2h} \Delta_0$. Summing the discounted geometric sequence gives the second expression in (17).

For the specialized stream, write $P_{T+h} = \text{diag}(p_{1,T+h}, p_{2,T+h})$. Diagonality is preserved because the state innovation covariance and the inherited covariance are diagonal and the report measures the first coordinate. Prediction followed by measurement gives

$$(A36) \quad s_{i,h} = \phi^2 p_{i,T+h-1} + q, \quad p_{1,T+h} = \frac{s_{1,h}}{1 + \tau s_{1,h}}, \quad p_{2,T+h} = s_{2,h}.$$

The second-coordinate recursion is affine with slope ϕ^2 , so it converges to $q/(1 - \phi^2)$ and differences between two inherited values equal ϕ^{2h} times their initial difference. For the first coordinate, define $f(p) = (\phi^2 p + q)/\{1 + \tau(\phi^2 p + q)\}$. On $[0, \infty)$, $0 \leq f'(p) = \phi^2/\{1 + \tau(\phi^2 p + q)\}^2 \leq \phi^2 < 1$. The contraction mapping theorem gives a unique fixed point and the first bound in (19). Solving $p = f(p)$ gives the positive root in (18); direct substitution gives the displayed value at $\phi = 0$. Posterior decision loss is $\text{tr}(DP)/2$. Since both covariance paths are diagonal and $D_{11}, D_{22} \geq 0$, applying (19) gives the final bound.

Appendix K. Posterior Construction from Bank Archives

K.1. Likelihood and Posterior

Consider retained reports $y_1, \dots, y_m$ at dates $s_1 < \dots < s_m$, with gaps $d_k = s_k - s_{k-1}$ and increments $z_k = y_k - y_{k-1}$. Conditional on the first report and a bank-specific pair (b, q) , the likelihood is proportional to $q^{-(m-1)/2} \exp\{-\sum_{k=2}^m (z_k - d_k b)^2 / (2q d_k)\}$. The level prior is diffuse, $b \mid q \sim N(0, q/\lambda_0)$, and $q \sim \text{IG}(a_0, \beta_0)$ under the density convention $q^{-a_0-1} \exp(-\beta_0/q)$. Put $\beta_0 = (a_0 - 1)q_0$, so the prior mean innovation variance is q_0 .

Let $S = s_m - s_1$, $Z = y_m - y_1$, and $U = \sum_{k=2}^m z_k^2 / d_k$. Completing the quadratic gives

$$(A37) \quad \lambda = \lambda_0 + S, \quad \widehat{b} = \frac{Z}{\lambda}, \quad a = a_0 + \frac{m-1}{2}, \quad \beta = \beta_0 + \frac{1}{2} \left(U - \frac{Z^2}{\lambda} \right).$$

The posterior is $b \mid q, H \sim N(\widehat{b}, q/\lambda)$ and $q \mid H \sim \text{IG}(a, \beta)$. Thus $\widehat{q} = \beta/(a - 1)$ and $\text{Var}(b \mid H) = \widehat{q}/\lambda$. For a target r quarters after the latest report, its predictive distribution has $2a$ degrees of freedom, location $y_m + r\widehat{b}$, and squared scale $(\beta/a)(r + r^2/\lambda)$. Its variance is (50) with $r = d_{it} + h$. The 90 percent intervals use the corresponding Student- t quantile.

At a current report, the level is known and its posterior variance is zero. Drift and future innovations retain positive variance. At age $d > 0$, the covariance of the current level-and-drift state is

$$(A38) \quad P_t = \widehat{q} \begin{pmatrix} d + d^2/\lambda & d/\lambda \\ d/\lambda & 1/\lambda \end{pmatrix}.$$

The transition matrix is $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, with conditional innovation covariance $q \text{diag}(1, 0)$. Equation (A38) describes a scale mixture of Gaussian posteriors. The full archive state also contains the posterior mean and inverse-gamma parameters. Conditional drift precision j^b remains defined when the current level has been observed exactly.

K.2. Expected Gain from an Intermediate Report

Before a current report is acquired, its predictive variance is $\widehat{q}(d + d^2/\lambda)$ and its covariance with the target h quarters later is $\widehat{q}\{d + d(d + h)/\lambda\}$. Its posterior-mean update is linear, with a coefficient independent of q . The law of total variance therefore gives expected MSE reduction

$$(A39) \quad \frac{\widehat{q}\{d + d(d + h)/\lambda\}^2}{d + d^2/\lambda} = \widehat{q} \left\{ d + h + \frac{(d + h)^2}{\lambda} - h - \frac{h^2}{\lambda + d} \right\}.$$

Integrating the conditional Gaussian calculation in Lemma 4 gives the same formula. Expected posterior q after the new report equals its current posterior expectation, and the updated drift precision factor is $\lambda + d$. At $d = 0$, the report repeats a known exact level and the gain is zero. The March and September semiannual archives have positive age.

K.3. Annual Prior Selection and Predictive Diagnostics

Prior selection uses $\lambda_0 \in \{4, 16, 64, 256, 1024\}$, $a_0 \in \{3, 6, 12\}$, and $q_0 \in \{0.25, 1, 4, 16, 64\}$. The 75 tuples are compared by mean negative predictive log density in year $y - 1$, averaged equally across available-twelve and semiannual-twelve archives. Ties use grid order. The selected tuple is held fixed for target year y , and bank posteriors use their retained origin records. Validation outcomes are dated at or before the first evaluation origin. Selected a_0 is 3 in every year. Selected q_0 is 1 in 2018 and 0.25 in the remaining years, at the lower grid boundary.

The model uses unbounded Gaussian increments for a bounded share. Predictive means are untrimmed in the posterior-state results so that loss and covariance calculations use the same specification. Coverage is evaluated from Student- t predictive intervals. Older histories can raise $\widehat{q}$ when they contain larger past changes. Pointwise posterior variance can consequently rise when those records enter, as in Table 3. Future-error diagnostics assess whether estimated dispersion corresponds to subsequent variation.

The same-current-report comparison uses main-sample rows with an observed origin ARM value and twelve available observations. Four-, eight-, and twelve-record histories retain the same current record. For variance calibration, origin-specific current-composition deciles and asset quintiles define joint cells. After subtracting cell means from squared error and predicted variance, ordinary least squares estimates their slope. Gain calibration uses latest semiannual composition deciles and asset quintiles within origin. Cell counts enter the residual degrees-of-freedom correction. Bank and target-date score totals give clustered slope variances. Negative estimated two-way variance is flagged and its interval left unreported.

Annual tuning and evaluation use chronological splits: prior parameters are selected from the preceding year, target-year posteriors use records available through each origin, and downstream forecasts use earlier target years. The diagnostic calculations use the same dated information sets.

Appendix L. Archive Score and the Report Budget

Under equal item costs and additive squared-error targets, the criterion maximizes $\sum_i a_i g_i$ over $a_i \in \{0, 1\}$ subject to $\sum_i a_i \leq k_t$. Its solution sorts g_i , with certificate number breaking ties. Uniform expected allocation assigns probability k_t/n_t to each eligible report within an origin. Heterogeneous monetary costs replace the constraint by $\sum_i c_i a_i \leq B_t$ and require the corresponding knapsack solution. Each evaluated allocation starts from its inherited semiannual history and holds the bank, realized report, and later outcome fixed across rules.

An accounting decomposition writes (51) as $\widehat{q}d + \widehat{q}\{(d+h)^2/\lambda - h^2/(\lambda+d)\}$. The first term is the value of observing accumulated innovations when drift is known; the second comes from uncertainty about drift. Both use the variance learned from the archive. Over the observed-origin sample, the second term accounts for 6.26 percent of total predicted gain. This decomposition explains why innovation scale provides a demanding benchmark for the mixed-state allocation rule.

Appendix M. Empirical Data and Sample Construction

The downloaded financial panel covers all 56 quarters from 2012 through 2025. Its 89 requested fields include identifiers and reporting metadata. Form 041 and 051 domestic commercial banks account for 268,189 source rows. The API field LNRERSF1 records adjustable-rate first-lien mortgages; LNRERSFM records the corresponding first-lien total. Monetary fields are in thousands of dollars. The six mortgage maturity or repricing buckets plus first-lien nonaccrual mortgages agree with the total within four thousand dollars throughout this source population. The comparison allows for rounding of separately reported entries.

The other-loan maturity reconciliation carries a flag for 11,436 source rows after including the gross-loan and unearned-income bridge. The strict sample excludes flagged origin or target observations. Aggregate interest-rate derivatives, RT, are included in common controls. Component derivatives fields have reporting gaps, so their values remain outside the input set. Scheduled zeros in the ARM item are retained when their total is positive. All March and September form-051 ARM entries are masked, including the 259 positive entries found in the domestic commercial source population.

Bank histories use dated FDIC certificates and RSSD identifiers. A merger, an identifier change, or a quarterly gap starts a new archive spell. Main-sample origins require three available ARM records, a June or December record, the common controls, and both future outcomes. The archive spell must continue through the income and ARM target dates. Quarterly flow reconciliation flags at the origin or either target and unresolved institution metadata are excluded. These target-date conditions describe the evaluated sample after outcomes are observed. The broad sample retains event and accounting flags subject to observed labels and origin controls; the strict sample applies the additional other-loan accounting filter.

The FRED inputs are the available daily observations of DFF, DGS2, and DGS10 within each quarter. Changes compare the current quarterly mean with its one- and four-quarter lags. The first four quarters use zero for an unavailable four-quarter rate change. All subsequent changes are dated before or at the forecast origin. The FDIC data use the August 28, 2026 download vintage.

Appendix N. Forecasting and Selection Algorithms

Let x_{it}^s denote the common and archive features for scenario s . The scenarios retain up to twelve available reports, twelve semiannual reports, one available report, four available reports, eight available reports, four reports after discarding one recent report, four after discarding two, and the semiannual archive with the current report restored. Training observations with shorter histories retain their available records and record the resulting

TABLE A1. Evaluation samples and target dates

Period	Sample	Rows	Banks	ARM dates	Income dates
2016	Main	19,480	5,070	2	4
	Other-loan check	18,374	4,995	2	4
	Broad	19,820	5,129	2	4
	Observed odd quarter	9,799	5,070	2	2
2022–2025	Main	59,892	4,096	8	16
	Other-loan check	55,349	4,087	8	16
	Broad	60,542	4,098	8	16
	Observed odd quarter	7,085	1,018	8	8
2016–2025	Main	165,049	5,294	20	40
	Other-loan check	148,956	5,282	20	40
	Broad	167,473	5,301	20	40
	Observed odd quarter	29,588	5,166	20	20

Rows are bank-origin observations with both targets. The main sample conditions on stable spells and the stated accounting checks. The broad sample retains event and accounting flags; the other-loan sample adds its reconciliation filter. Observed odd quarters are March and September origins with an actual scheduled ARM value.

count. The dedicated record-length and recency evaluations require the specified counts in every compared scenario.

For each outcome, the ridge coefficient minimizes

$$(A40) \quad \frac{1}{8N_{\text{train}}} \sum_{s=1}^8 \sum_{it \in \text{train}} (Y_{it} - a - x_{it}^s b)^2 + \lambda b' b.$$

The intercept is unpenalized. Financial ratios and current and lagged income features are clipped at their training 0.5 and 99.5 percentiles. Missing lagged entries use training medians with a lag-availability indicator. Features are standardized using training means and standard deviations. Rate levels and changes retain their observed ranges, and responses retain their observed values. The penalty grid is $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$. Validation selects the smallest average MSE over the available-report and semiannual scenarios, with equal weights. Ties select the larger penalty. The selected penalty is used after re-estimation and preprocessing on all preceding target years. ARM predictions are restricted to $[0, 100]$.

Common controls contain log assets; mortgage, capital, deposit, securities, loan, nonaccrual, and aggregate interest-rate derivative balances scaled by assets; six mortgage maturity shares; six other-loan maturity balances; eight deposit maturity balances; twelve securities maturity balances; current and lagged income and its change; current rate levels and dated changes; the Treasury slope; quarter indicators; and the known ARM horizon.

ARM archives contribute the latest percentage share, mean, standard deviation, age, count, span, and slope. The latest and average share interact with the current mortgage-to-asset ratio, and the latest exposure also interacts with the federal funds rate and its quarterly change.

The persistence forecast uses the last observed ARM share and current quarterly net interest income divided by current assets. The Gaussian comparator first regresses observed current ARM shares on common controls, using a fixed ridge penalty of 0.01. It filters residuals with a scalar random-walk state and temporary observation variation. The state-innovation to observation-variance ratio is selected from $\{0.001, 0.01, 0.1, 1, 10\}$ using validation forecast MSE, averaged over the two schedules. Prior residual variance is ten observation-variance units. Each retained archive is processed in date order, and gaps enter the state-variance update. The ARM forecast combines the filtered residual with the current common-control prediction and is clipped to $[0, 100]$. A separate income ridge regression uses the common controls and this filtered exposure. Its penalty uses the same validation grid. This is a statistical forecasting approximation; the residual observation variance represents temporary composition variation.

The auxiliary mortgage-total forecast uses common controls and a validation-selected ridge penalty. Its target is next-quarter first-lien mortgages divided by origin assets in basis points. Its input set is common to the reporting scenarios, so the schedule exercise supplies the same total forecast in each scenario.

For allocation, define validation gain $g_{it} = (Y_{it} - \hat{f}_{it}^S)^2 - (Y_{it} - \hat{f}_{it}^{S+})^2$. A ridge regression with penalty 0.1 projects g_{it} on the semiannual feature vector in the preceding validation year, using that year's preprocessing. The score is fitted separately for each target. Selection ranks the resulting score within the evaluation origin, breaking ties by certificate. Asset selection uses the same tie breaker. A fixed seed generates uniform draws. For k reports and n eligible banks, the expected uniform loss is $n^{-1} \sum_i [\ell_i^S + (k/n)(\ell_i^{S+} - \ell_i^S)]$. This expression averages the realized losses over all uniform subsets of size k .

Appendix O. Empirical Uncertainty and Additional Results

Prediction intervals use the preceding validation year's 90th percentile of absolute errors for each procedure and reporting scenario. ARM endpoints are clipped to the feasible share range. Reported coverage is the fraction of subsequent realizations inside those intervals. The serially dependent panel and annual distribution changes make this an empirical calibration exercise. Interval radii are empirical residual quantiles of fitted procedures. The calculations condition on estimated parameters.

Sampling summaries use paired squared-error differences. The bank-cluster variance sums squared bank totals of centered differences, with the cluster-count correction. Holding-company sensitivity groups certificates using the transitive union of their ob-

TABLE A2. Paired squared-error differences and interval calculations

Target	Difference	Bank 95%	Holder 95%	Two-way 95%
<i>2016</i>				
ARM	8.64	[5.84, 11.45]	[5.84, 11.45]	[-2.49, 19.78]
Income	0.17	[0.10, 0.24]	[0.10, 0.24]	[-0.01, 0.35]
<i>2022–2025, observed odd quarters</i>				
ARM	21.04	[14.28, 27.80]	[14.26, 27.82]	[11.57, 30.51]
Income	0.08	[-0.23, 0.39]	[-0.24, 0.40]	[-1.25, 1.41]
<i>2016–2025, observed odd quarters</i>				
ARM	18.90	[15.83, 21.97]	[15.66, 22.14]	[15.81, 21.99]
Income	0.06	[-0.06, 0.19]	[-0.06, 0.19]	[-0.25, 0.38]

Differences equal semiannual loss minus available-report loss for the ridge procedure. Units are squared percentage points for ARM and squared basis points for income. Bank and holder intervals condition on the observed dates. Two-way intervals add target-quarter clustering; the smaller cluster count determines the critical value. The 2016 ARM calculation uses two target-quarter clusters.

TABLE A3. Empirical coverage of 90 percent forecast intervals

Procedure	ARM semi.	ARM avail.	Income semi.	Income avail.
<i>2016</i>				
Ridge	89.5	89.2	47.7	47.7
Persistence	90.6	90.2	91.3	91.3
Gaussian filter	88.8	90.3	28.3	28.3
<i>2022–2025, observed odd quarters</i>				
Ridge	92.6	96.1	92.5	92.8
Persistence	92.6	96.1	94.4	94.4
Gaussian filter	90.6	96.5	92.6	92.7

Entries are percentages of subsequent outcomes within intervals calibrated on the preceding validation year. ARM intervals respect the share bounds. Coverage is an evaluated property of these intervals.

served historical holding-company memberships. These groups enter uncertainty calculations only. The two-way calculation adds bank and target-quarter cluster components and subtracts their intersection component, following Cameron et al. (2011). Its t critical value uses the smaller cluster count minus one. Negative two-way variance estimates are omitted from interval reporting. In 2016, the two ARM target dates and four income target dates constrain this approximation. Year-specific differences and the count of target dates accompany pooled results.

Appendix P. Predictive Diagnostics

For test year Y , tuning and calibration use target year $Y - 1$, and fitting labels end by the first origin of Y . The specifications below examine predictive variance, bounded-share transformations, local scale, and repeated reported values under this chronological rule.

The expanded grid uses $\lambda_0 \in \{4, 16, 64, 256, 1024\}$, $a_0 \in \{1.1, 1.5, 2, 3, 6, 12\}$, and $q_0 \in \{0.0001, 0.001, 0.01, 0.025, 0.0625, 0.125, 0.25, 1, 4, 16, 64\}$. The criterion averages Student- t negative log density over the available and semiannual histories. Predictive second moments exist because $a_0 > 1$. The complete profiles include constant-history and varying-history subsets, with the latter diagnostic computed on the preceding-year validation sample. A constant history followed by the same value receives increasing log density as the scale approaches zero. Constant histories followed by a change generate the opposite pressure. Their relative frequencies help determine the observed boundary selection. Appendix P.3 estimates a likelihood for discrete persistence and continuous changes at the reporting resolution.

The variance-floor fit solves $\min_{\alpha, \beta \geq 0} \sum_i \{(Y_i - \hat{m}_i)^2 - \alpha - \beta V_i\}^2$ on the preceding year's available and semiannual histories. Test-year means use the baseline posterior, and predictive variance is $V_i^c = \alpha + \beta V_i$. A Student- t_5 distribution centered at that mean is scaled to variance V_i^c . The acquisition proxy is $g_i V_i^c / V_i$, evaluated on the semiannual archive. This calibration supplies a predictive variance and a ranking rule. Its coefficients are summaries of prior forecast errors.

The bounded specification transforms a share y in percentage points to $z = \log\{(y + \epsilon)/(100 - y + \epsilon)\}$ with $\epsilon = 0.1$. It fits the same conjugate increment model on z , using the displayed λ_0, a_0 grids and $q_0 \in \{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 0.1, 1, 4\}$. The inverse map is $\max\{0, \min\{100, (100 + 2\epsilon) \logit^{-1}(z) - \epsilon\}\}$. Predictive moments use 513 Gauss-Legendre nodes in probability space. Comparisons with 1,025 nodes give maximum differences below 0.000063 percentage points in means and 0.019 squared points in variances. Its acquisition proxy multiplies the latent Gaussian gain by the squared derivative of the smooth inverse map at the latent predictive mean. This is a delta-method ranking, evaluated by realized forecast loss.

The local-scale benchmark centers its prediction on the last retained share. For retained increments Δy_r over gaps d_r , let $w_r = 2^{-\ell_r/L}$, where ℓ_r is the age of the newer end-point relative to the latest retained record. Its variance scale is $\tilde{q} = (\nu \bar{q} + \sum_r w_r \Delta y_r^2 / d_r) / (\nu + \sum_r w_r)$. The pooled $\bar{q}$ averages historical squared increments in preceding-year available and semiannual windows. The half-life $L \in \{1, 2, 4, 8\}$ and shrinkage weight $\nu \in \{1, 4, 16\}$ minimize preceding-year Student- t_5 negative log density. Predictive variance is $(d + h)\tilde{q}$, and the acquisition proxy is $d\tilde{q}$. This is a heavy-tail local-scale predictive specification. A latent stochastic-volatility transition law would introduce additional parameters.

TABLE A4. Predictive calibration under alternative specifications

Specification	RMSE	Mean variance	Coverage (%)	Q1 variance	Q1 MSE
Baseline posterior	7.804	45.18	85.6	0.167	33.46
Expanded prior grid	7.808	60.34	87.0	0.009	33.46
Variance floor	7.804	67.12	96.4	45.430	33.46
Bounded transformation	7.909	76.89	87.1	0.032	33.45
Local scale, t_5	7.714	60.31	93.7	8.433	33.46

The 101,921 current-report origins are shared across specifications. Q1 fixes the lowest within-origin variance quintile under the baseline posterior (20,360 observations). Means and variances use percentage-point units; coverage concerns 90 percent predictive intervals. Each annual fit uses the preceding year. The variance floor calibrates squared errors; the local-scale benchmark uses a Student- t distribution with five degrees of freedom. These distributions are evaluated as predictive specifications.

TABLE A5. Reported-value resolution and the mixed predictive state

Band (bp)	Observed share	Predicted share	AUC	MSE/V	Both balances equal
0	19.50	19.46	0.996	0.97	0.06
1	20.61	20.56	0.989	0.97	0.06
5	23.27	23.20	0.968	0.97	0.06
10	25.63	25.45	0.950	0.97	0.06

Observed and predicted shares are percentages in the 2017–2025 out-of-time sample. A band of one basis point equals 0.01 percentage point of mortgage composition. Each row redefines the low-change component, refits the expanding-window classifier, and recomputes mixture variance. The last column reports exact equality of the ARM balance and mortgage-total balance between the latest retained report and the target report; it is repeated across band rows because its definition uses raw balances.

Gradient boosting uses a HistGradientBoostingRegressor with squared loss, 120 iterations, learning rate 0.05, 15 maximum leaves, minimum leaf size 100, and L_2 penalty 1. Early stopping is disabled. Training stacks the available and semiannual histories with weight one half each and uses the current-bank and archive features. Training labels end in year $Y - 1$. Native missing-value handling is used, and share forecasts are clipped to $[0, 100]$. The hyperparameters are fixed across years. Persistence uses the last available report. For observed March/September origins, acquisition supplies the same latest record as available access.

P.1. The Common Mixed-State Acquisition Score

The score uses the estimated state for the intermediate report, semiannual archive features, and common bank controls available at the origin. Its label is the equal-weight mean of paired squared-error reductions under ridge, persistence, and gradient boosting in (52). Candidate boosting regressions have configurations (leaves, minimum leaf, L_2)

equal to (7, 250, 10), (15, 200, 20), (15, 250, 40), (7, 500, 40), and (31, 100, 10). The learning rate is 0.04, each fit has 220 iterations, and labels are winsorized at their 1st and 99th training percentiles.

The 2018 architecture is selected by five-fold bank cross-validation within target year 2017. For evaluation year $Y \geq 2019$, each candidate is fitted through $Y - 2$ and its acquisition value is measured in $Y - 1$; the winner is refitted through $Y - 1$ and applied in Y . Ties use the listed order. Each annual score is fixed before its evaluation outcomes. At budget b , the rule selects $\lfloor bn_t \rfloor$ reports within origin t , with certificate number breaking score ties.

For each budget, the paired value contrast uses the same bank, realized intermediate report, and later outcome under both allocation rules. Let $\widehat{\Sigma}_B$, $\widehat{\Sigma}_T$, and $\widehat{\Sigma}_{B \cap T}$ denote cluster-score covariance matrices by bank, target quarter, and their intersection. The budget-family covariance is $\widehat{\Sigma} = \widehat{\Sigma}_B + \widehat{\Sigma}_T - \widehat{\Sigma}_{B \cap T}$. A Gaussian maximum statistic over the 10, 25, and 50 percent budgets supplies simultaneous intervals, with its critical value multiplied by the ratio of the t_{15} and standard-normal 97.5 percent quantiles. Sixteen leave-one-target-quarter estimates measure time support. The main table compares the sequential score with uniform expected allocation and innovation-scale ranking. The three-budget grid was declared after the earlier 25 percent result; simultaneous inference covers emphasis within the displayed grid.

P.2. The Limit of the Continuous-Density Score

Let $U_{is} = \sum_r (\Delta y_r)^2 / d_r$ for the retained increments of scenario s . For a validation observation i whose retained increments are all zero, write $a_i = a_0 + (n_i - 1)/2$. Its posterior mean is the last retained share and its predictive scale squared is proportional to q_0 . If the outcome repeats that share, its negative log density has coefficient 1/2 on $\log q_0$. If the outcome changes, the Student- t tail gives coefficient $-a_i$. Histories with a positive squared increment have a positive limiting posterior scale for fixed $\lambda_0 > 0$. Thus, averaging equally over the two access scenarios s and N_Y validation rows,

$$(A41) \quad c_Y = \frac{1}{2N_Y} \sum_{s,i: U_{is}=0} \left\{ \frac{1}{2} \mathbf{1}(Y_i = y_{is,\text{last}}) - a_{is} \mathbf{1}(Y_i \neq y_{is,\text{last}}) \right\}, \quad \text{NLL}(q_0) = c_Y \log q_0 + O(1).$$

At $a_0 = 1.1$, the ten annual coefficients range from 0.03169 to 0.06634; at $a_0 = 3$, they range from 0.01309 to 0.05379. Their signs imply an unbounded criterion as q_0 approaches zero. This result explains the boundary behavior of the continuous-density tuning criterion. Appendix P.3 uses a mixed likelihood for the predictive state. The fixed-grid score remains a benchmark input to the common acquisition rule.

P.3. A Point-Mass and Change Likelihood

Let $R_{it} = 1$ when the target share equals the latest share in the semiannual archive at the numerical precision of the constructed reported percentage, and let $p_{it} = \Pr(R_{it} = 1 \mid X_{it})$. Conditional on a change, the target has a Student- t_5 density $f_5(y; m_{it}, v_{it})$ centered on the expanded-grid predictive mean. The likelihood contribution relative to the sum of a point mass at the latest share and Lebesgue measure is

$$(A42) \quad \mathcal{L}_{it} = p_{it}^{R_{it}} \{(1 - p_{it})f_5(Y_{it}; m_{it}, v_{it})\}^{1-R_{it}}.$$

The probability model is a histogram gradient-boosting classifier with learning rate 0.05, 120 iterations, at most 15 leaves, minimum leaf size 150, and L_2 penalty one. Its inputs are the log baseline variance, absolute drift, log innovation estimate, record count, age, span, latest share, predicted change, log assets, calendar quarter, Call Report form, forecast horizon, origin date, and mortgage-to-asset ratio. The feature set excludes the current ARM share under semiannual access. For target year Y , all target years before Y form the training sample. The first prediction year is 2017.

The conditional change variance uses twenty cells defined by five-percent quantiles of the expanded-grid variance among training observations with $R_{it} = 0$. Within a cell, v_{it} equals the training mean of $(Y_{it} - m_{it})^2$. Test observations use the training cutoffs and cell variances. The mixture mean and variance are

$$(A43) \quad \mu_{it} = p_{it}\ell_{it} + (1 - p_{it})m_{it}, \quad V_{it} = (1 - p_{it})v_{it} + p_{it}(1 - p_{it})(\ell_{it} - m_{it})^2,$$

where ℓ_{it} is the latest retained share. Quantiles invert the continuous t_5 distribution on each side of its jump of size p_{it} at ℓ_{it} . The mixed log-score comparison in the text changes p_{it} to the expanding-window repeat frequency and keeps f_5 fixed, so both scores use the measure in (A42).

For the resolution sensitivity, $R_{it}(b) = 1\{|Y_{it} - \ell_{it}| \leq b/100\}$ for $b \in \{0, 1, 5, 10\}$ basis points of composition. Each definition refits the annual classifier, re-estimates the conditional change-variance cells, and recomputes mixture moments. The accounting comparison uses the ARM numerator and first-lien mortgage denominator for the target report and the latest retained report. Exact share repetition occurs in 19.76 percent of the full evaluated sample; the ARM numerator is equal in 19.01 percent, the denominator is equal in 0.08 percent, and both are equal in 0.06 percent. At the intermediate report origin, the corresponding shares are 59.81, 59.75, 49.92, and 49.89 percent. These counts distinguish a repeated ratio from repeated accounting inputs.

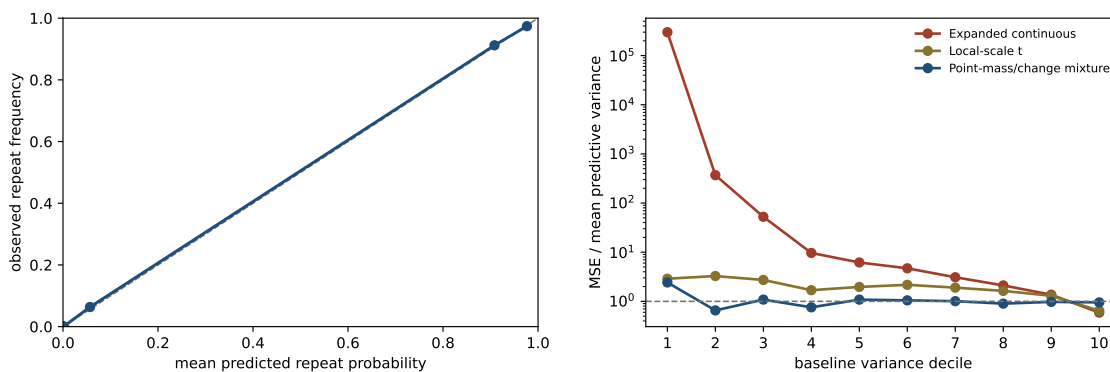


FIGURE A2. Repeat probabilities and predictive-variance calibration

The left panel groups the 2017–25 out-of-time observations into deciles of predicted repeat probability. The right panel groups them into deciles of the baseline semiannual variance and reports realized MSE divided by mean predictive variance. The horizontal line marks equality. Each annual fit uses earlier target years.

Appendix Q. Reform Cohorts and Eligibility

The cohort construction fixes domestic commercial-bank cohorts at June 2016 and June 2018, using bank classes N, NM, and SM and forms 041 or 051. The first cohort contains 4,654 banks below one billion dollars; the second contains 455 banks with assets from one billion to below five billion dollars. Subsequent presence is measured in the same bank-class/form sample. These are certificate-based cohort histories; merger exits and changes in bank classification alter the present denominator.

The 2019 eligibility rule uses preceding-June assets and also imposes conditions concerning foreign offices, advanced approaches, and institution classification, with supervisory reservation of authority (Board of Governors of the Federal Reserve System, Federal Deposit Insurance Corporation, and Office of the Comptroller of the Currency 2019). The asset/class/form criteria in this cohort construction are observable proxies. Voluntary adoption can depend on expected growth and the costs of maintaining an existing reporting system. The 2017 one-billion-dollar boundary also coincides with examination-cycle rules associated with the same asset boundary. A causal use of eligibility would require a first-stage analysis, exclusion restrictions addressing concurrent changes, and a treatment of selection into continued observation. The 2019 ten-percent window has zero observed 051 adoption on either side in September 2019 and March 2020.

Appendix R. Income Ablations

The same records can be evaluated against next-quarter net interest income. A shared ridge coefficient vector for each outcome takes current bank characteristics, maturity measurements, earnings, rates, and archive summaries as inputs. Coefficients are shared

TABLE A6. Income forecast information and the marginal ARM record

Information set	Semiannual RMSE	Available RMSE	MSE gain (%)	MSE reduction
2016: main sample; $N = 19,480$				
All controls	10.778	10.770	0.146	0.170
Earnings removed	16.024	15.975	0.600	1.539
Maturity removed	10.832	10.829	0.052	0.061
Both groups removed	14.922	14.889	0.444	0.988
2022–2025: observed March/September; $N = 7,085$				
All controls	11.069	11.066	0.065	0.080
Earnings removed	21.620	21.632	-0.106	-0.497
Maturity removed	11.042	11.037	0.089	0.108
Both groups removed	21.500	21.513	-0.118	-0.546
2016–2025: observed March/September; $N = 29,588$				
All controls	11.623	11.620	0.047	0.064
Earnings removed	18.473	18.456	0.182	0.622
Maturity removed	11.684	11.681	0.038	0.051
Both groups removed	18.179	18.165	0.149	0.494

The shared eight-scenario ridge procedure and rolling samples coincide across restrictions. Earnings comprise four current/lag/change/availability features. Maturity comprises six mortgage, six other-loan, eight deposit, and twelve securities bins. Remaining balance sheets, rates, and archive inputs stay available. RMSE is in basis points of origin assets; MSE reductions are in squared basis points. Positive gains favor the available-report archive.

across eight access restrictions. For year y , fitting uses outcomes through $y - 2$, penalty selection uses $y - 1$, and refitting uses outcomes through $y - 1$. Comparisons use identical bank-origin rows. Appendix N gives the procedure and its comparators.

In the 2016 schedule comparison, quarterly access reduces composition MSE by 17.02 percent and income MSE by 0.15 percent. Table A6 removes the earnings group, the maturity group, and both groups while preserving the fitting rule, remaining predictors, and sample. Earnings comprise current income, its lag, its change, and the lag-availability indicator. Maturity comprises six mortgage, six other-loan, eight deposit, and twelve securities measurements. These restrictions evaluate marginal ARM record value under different existing income information. Removing earnings raises available-archive income RMSE from 10.770 to 15.975 basis points in 2016, and from 11.066 to 21.632 in the observed March and September sample during 2022–2025. Its effect on the marginal ARM record varies across those periods. The income MSE gain rises from 0.146 to 0.600 percent in 2016 and changes from 0.065 to -0.106 percent in 2022–2025. In the pooled observed-origin sample, the increase in the paired ARM gain after removing earnings is 0.558 squared basis points, with two-way interval $[-0.347, 1.463]$.

Current earnings reduce overall income forecast error in these comparisons. The ARM contribution stays small and varies with the period and remaining predictors. This target dependence corresponds to the covariance term in Lemma 4; the reported ablations estimate forecast losses under the specified procedures. A policy-loss matrix would additionally require the exposure of policy actions to those income outcomes. The 2016 persistence income forecast has RMSE 5.761 basis points, compared with 10.770 for full-information ridge. Its 90 percent interval coverage is 91.3 percent; ridge coverage is 47.7 percent. Appendix O reports forecast errors and coverage for every procedure. The ablations describe a fitted income forecast’s response to its information set, including its estimation error.

References

- Board of Governors of the Federal Reserve System, Federal Deposit Insurance Corporation, and Office of the Comptroller of the Currency. Reduced reporting for covered depository institutions, 2019. URL <https://www.federalreserve.gov/newsevents/pressreleases/files/bcreg20190617a1.pdf>. Final rule released June 17, 2019; reporting changes effective September 30, 2019.
- A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Robust inference with multiway clustering. *Journal of Business & Economic Statistics*, 29(2):238–249, 2011. doi: 10.1198/jbes.2010.07136.